

SYDNEY TECHNICAL HIGH SCHOOL



HIGHER SCHOOL CERTIFICATE ASSESSMENT TASK 3

JUNE 2016

Mathematics Extension 2

General Instructions

- Working time - 90 minutes
- Write using black pen only
- Board-approved calculators may be used
- All necessary working should be shown in questions 6 to 9
- Start each question on a new page
- A Board of Studies reference sheet is provided

Total marks - 60

Section 1 - 5 marks

Attempt Questions 1 – 5.
Allow about 8 minutes for this section.

Section 2 - 55 marks

Attempt Questions 6 – 9.
Allow about 82 minutes for this section.

Name : _____

Teacher : _____

Section 1 (5 marks)

Attempt Questions 1 – 5

Use the multiple-choice answer sheet in your answer booklet for Questions 1 – 5.
Do not remove the multiple-choice answer sheet from your answer booklet.

1. Which of the following is equivalent $\int x \sec^2(x^2) dx$?

- (A) $2 \tan(x^2) + c$
- (B) $\frac{1}{2} \tan(x^2) + c$
- (C) $\frac{1}{6} \sec^3(x^2) + c$
- (D) $\frac{1}{3} \sec^3(x^2) + c$

2. What is the multiplicity of the root $x = -1$ of the equation

$$3x^5 - 5x^4 - 35x - 27 = 0 \text{ ?}$$

- (A) 1
- (B) 2
- (C) 3
- (D) 4

3. If $y = \cos^{-1}(e^x)$ then $\frac{dy}{dx}$ equals

- (A) $-\operatorname{cosec} y$
- (B) $-\tan y$
- (C) $-\cot y$
- (D) $-\sec y$

4. The equation $2x^3 - 7x + 1 = 0$ has roots α , β and γ .

What is the value of $\alpha^3 + \beta^3 + \gamma^3$?

(A) 0

(B) $\frac{43}{4}$

(C) $-\frac{1}{2}$

(D) $-\frac{3}{2}$

5. Which integral has the smallest value?

(A) $\int_0^{\frac{\pi}{4}} \sin^2 x \, dx$

(B) $\int_0^{\frac{\pi}{4}} \cos^2 x \, dx$

(C) $\int_0^{\frac{\pi}{4}} \sin x \cos x \, dx$

(D) $\int_0^{\frac{\pi}{4}} \sin x \tan x \, dx$

Section 2 (55 marks)

Attempt Questions 6 – 9

Start each question on a new page

Question 6 (14 marks)

a) Find $\int \frac{dx}{\sqrt{6x-x^2}}$ 2

b) Evaluate $\int_0^{\sqrt{3}} \frac{x^3}{(1+x^2)^{\frac{3}{2}}} dx$ using the substitution $u = 1 + x^2$ 3

c) Use integration by parts to evaluate $\int_1^e \frac{\ln x}{\sqrt{x}} dx$ 3

d) If α , β and γ are the roots of the equation $x^3 - 5x - 3 = 0$ 3

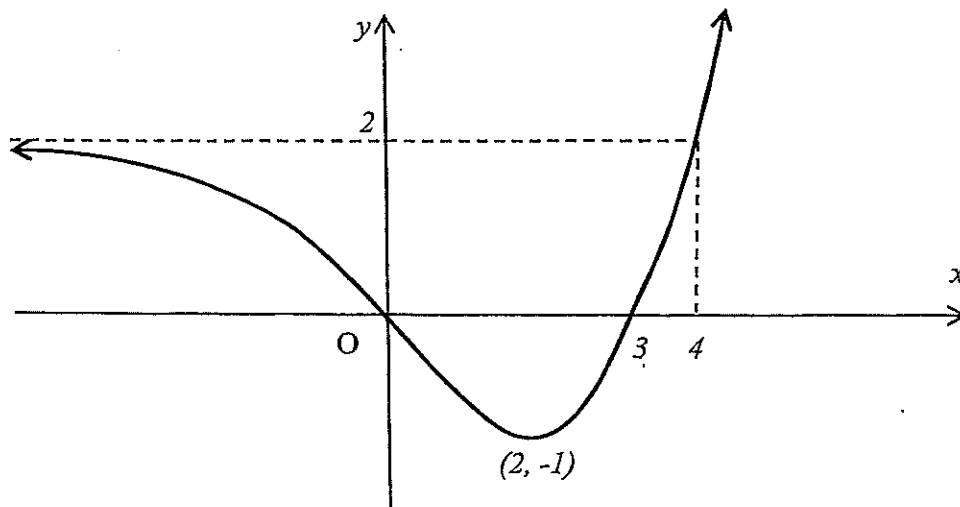
find the monic polynomial equation whose roots are $\frac{\alpha}{\beta\gamma}$, $\frac{\beta}{\alpha\gamma}$ and $\frac{\gamma}{\alpha\beta}$

e) Find the real values of a and b given that $3 + i$ is a root 3

of the equation $z^3 + az^2 + bz + 10 = 0$

Question 7 (13 marks) (Start a new page)

- a) The diagram below shows the graph of $y = f(x)$.



Draw neat sketches, on separate diagrams, of the following;

- | | |
|--|---|
| i) $y = f(-x)$ | 2 |
| ii) $y = \frac{1}{f(x)}$ | 2 |
| iii) $y = [f(x)]^3$ | 2 |
| b) Use the substitution $t = \tan \frac{\theta}{2}$ to find $\int \frac{\tan \theta}{1 + \cos \theta} d\theta$ | 3 |
| c) For what values of k does the equation $x^3 - 3x^2 - 24x + k = 0$ have exactly one real root? | 4 |

Question 8 (14 marks) (start a new page)

- a) Without the use of calculus, sketch the curve $y^2 = x^2(4 - x^2)$ 2
- b) Find $\int \frac{x+6}{x^2+4x+29} dx$ 4
- c) If the polynomial $ax^{n+1} + bx^n + 1$ is divisible by $(x - 1)^2$, 3
find expressions for a and b in terms of n .
- d) i) If $I_n = \int_0^1 x^n e^{x^2} dx$ show that $I_n = \frac{e}{2} - \left(\frac{n-1}{2}\right) I_{n-2}$ 3
- ii) Evaluate $\int_0^1 x^5 e^{x^2} dx$ 2

Question 9 (14 marks) (start a new page)

- a) Use partial fractions to find $\int \frac{x-3}{x^2+6x+5} dx$ 3
- b) Find $\int \cos^5 x dx$ 3
- c) i) Solve $z^5 - 1 = 0$ over the complex field. 2
ii) Considering the sum of the roots found in part i), or otherwise,
determine the exact value of $\cos \frac{2\pi}{5}$. 2
- d) The equation $x^3 + ax^2 + bx + c = 0$ 4
has one root equal to the sum of the other two roots.
Show that $a^3 - 4ab + 8c = 0$.

End of Paper

B 2. B 3. C 4. D 5. A

a) $\int \frac{dx}{\sqrt{9-(x-2)^2}}$

$= \sin^{-1} \frac{x-2}{3} + C$

b) $\int_0^3 \frac{x^3}{(4+x^2)^2} dx$ $u=4+x^2$
 $du=2x dx$

$= \frac{1}{2} \int_0^3 \frac{x^2 \cdot 2x dx}{(4+x^2)^2}$

$= \frac{1}{2} \int_4^7 \frac{u-4}{u^2} du$

$= \frac{1}{2} \int_4^7 \left(\frac{1}{u} - \frac{4}{u^2} \right) du$

$= \frac{1}{2} \left[\ln u + \frac{4}{u} \right]_4^7$

$= \frac{1}{2} \left[(\ln 7 + 1) - (\ln 4 + 1) \right]$

$= \frac{1}{2} \ln \frac{7}{4}$

c) $\int_1^e \frac{e^{\ln x} dx}{\sqrt{x}}$ $u=\ln x$ $V=2\sqrt{x}$
 $u'=\frac{1}{x}$ $V'=\frac{1}{\sqrt{x}}$

$= 2\sqrt{x} \ln x - \int_1^e 2 dx$

$= 2\sqrt{e} - [4x]_1^e$

$= 2\sqrt{e} - 4\sqrt{e} + 4$

$= 4 - 2\sqrt{e}$

$\frac{dx}{dt} = \frac{d^2}{dt^2}$

$= \frac{d^2}{dt^2}$

$\therefore$ roots $\frac{d^2}{dt^2}, \frac{d}{dt}, \frac{d^2}{dt^2}$

replace x with $\sqrt{3}x$

$\therefore (\sqrt{3}x)^3 - 5\sqrt{3}x - 3 = 0$

$3x\sqrt{3}x - 5\sqrt{3}x = 3$

$(3x\sqrt{3}x - 5\sqrt{3}x)^2 = 9$

$27x^3 - 90x^2 + 75x - 9 = 0$

$x^3 - \frac{10}{3}x^2 + \frac{25}{9}x - \frac{1}{3} = 0$

e) roots $3+i, 3-i, \alpha$

$\therefore (3+i)(3-i)\alpha = -10$

$10\alpha = -10$
 $\alpha = -1$

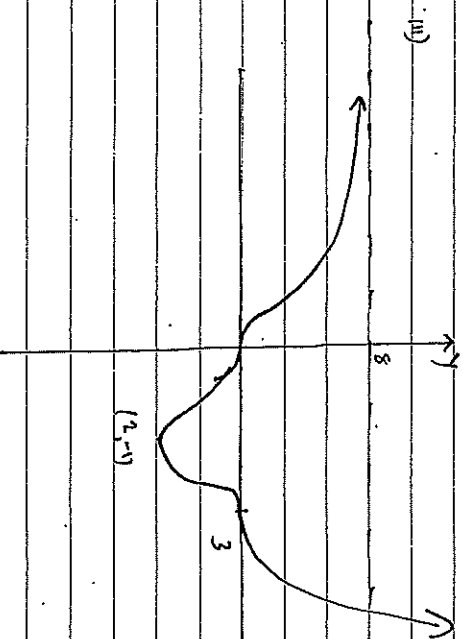
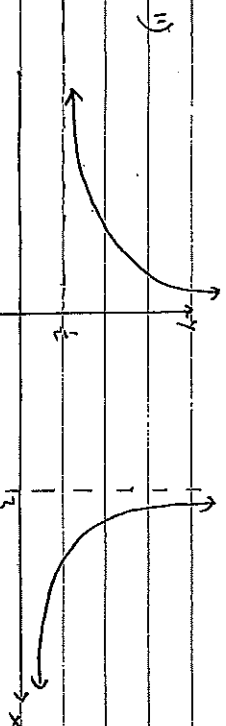
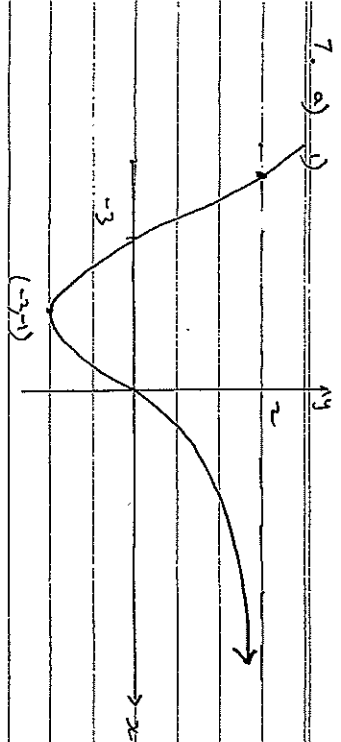
$\therefore 3+i+3-i-1 = -\alpha$

$\alpha = -5$

sub $x = -1$

$-1-5-b+10=0$
 $b=4$

$\therefore \alpha = -5, b=4$



$$\int \frac{2x}{(1+x^2)(1+x^2)} \cdot \frac{2x}{(1+x^2)} dx$$

$$= \int \frac{\frac{4x}{1+x^2} dx}{(1+x^2)(1+x^2)}$$

$$= \int \frac{4x}{2(1+x^2)^2} dx$$

$$= \int \frac{-2x}{1+x^2} dx$$

$$= -\ln(1+x^2) + C$$

2. $-\ln(1+x^2) + C$

c) St 1 $3x^2 - 6x - 24 = 0$

$$3(x^2 - 2x - 8) = 0$$

$$3(x-4)(x+2) = 0$$

1. $3x^2 - 6x - 24 = 0$

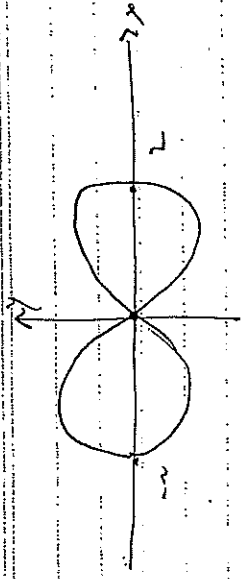
$$x^2 - 2x - 8 = 0$$

$$(x-4)(x+2) = 0$$

$x = 4$ $y = -8 - 12 + 48 + 4 = 22$

$\therefore (k-50)(25+k) > 0$

$$k > 80, k < -28$$



b.)

$$\begin{aligned}
 & b) \int \frac{x+6}{x^2+4x+29} dx \\
 & = \frac{1}{2} \int \frac{2x+4}{x^2+4x+29} dx + \int \frac{4}{x^2+4x+29} dx \\
 & = \frac{1}{2} \ln |x^2+4x+29| + \int \frac{4}{(x+2)^2+25} dx \\
 & = \frac{1}{2} \ln |x^2+4x+29| + \frac{4}{5} \tan^{-1}\left(\frac{x+2}{5}\right) + C
 \end{aligned}$$

c.) 1 is a double root

$$\begin{aligned}
 & (n+1)ax^n + nbx^{n-1} = 0 \text{ with } n=1 \\
 & a(n+1) + nb = 0
 \end{aligned}$$

$$a + b + 1 = 0$$

$$b = -a - 1$$

$$\begin{aligned}
 & a(n+1) \neq n(a+1) = 0 \\
 & a(n+1) - n(a+1) = 0
 \end{aligned}$$

$$a = 0$$

$$b = -1$$

$$d) i) I_n = \int_0^1 x^n e^x dx$$

$$= \frac{1}{n} \int_0^1 x^{n-1} e^x dx$$

$$\begin{aligned}
 & u = x^{n-1} \quad v = e^x \\
 & u' = (n-1)x^{n-2} \quad v' = e^x
 \end{aligned}$$

$$= \frac{1}{n} \left[x^{n-1} e^x \right]_0^1 - \int_0^1 (n-1)x^{n-2} e^x dx$$

$$I_n = \frac{1}{n} e - (n-1) I_{n-1}$$

$$ii) I_3 = \frac{1}{3} e - 2 I_2$$

$$= \frac{1}{3} e - 2 \left[\frac{1}{2} e - I_1 \right]$$

$$= \frac{1}{3} e - e + 2 \int_0^1 x e^x dx$$

$$= \frac{1}{3} e - e + [x e^x]_0^1$$

$$= \frac{1}{3} e - e + e$$

$$= \frac{1}{3} e$$

$$9. \frac{x-3}{(x+5)(x+1)} = \frac{A}{x+5} + \frac{B}{x+1}$$

$$a) (x+5)(x+1) \quad x+5 \quad x+1$$

$$\therefore A(x+1) + B(x+5) = x-3$$

$$\text{let } x = -1 \quad B = -1$$

$$\text{let } x = -5 \quad A = 2$$

$$\therefore \int \frac{2}{x+5} - \frac{1}{x+1} dx$$

$$= 2 \ln|x+5| - \ln|x+1| + C$$

$$b) \int \cos^5 x dx$$

$$= \int \cos x \cos^4 x dx$$

$$= \int \cos x (1 - \sin^2 x)^2 dx \quad \text{let } u = \sin x$$

$$= \int (1 - u^2)^2 du \quad du = \cos x dx$$

$$= \int (1 - 2u^2 + u^4) du$$

$$= u - \frac{2}{3} u^3 + \frac{1}{5} u^5$$

$$= \sin x - \frac{2}{3} \sin^3 x + \frac{1}{5} \sin^5 x + C$$

$$c) 1) \text{ Solutions are } z_1 = 1$$

$$z_2 = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \quad \text{these can be}$$

$$z_3 = \cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \quad \text{written in}$$

$$z_4 = \cos \frac{4\pi}{3} - i \sin \frac{4\pi}{3} \quad \text{very different}$$

$$z_5 = \cos \frac{2\pi}{3} - i \sin \frac{2\pi}{3} \quad \text{very}$$

$$11) z_1 + z_2 + z_3 + z_4 + z_5 = 0$$

$$\therefore 2 \cos \frac{2\pi}{3} + 2 \cos \frac{4\pi}{3} = -1$$

$$\cos \frac{2\pi}{3} + (2 \cos^2 \frac{2\pi}{3} - 1) = -\frac{1}{2} \quad \text{double angle w/d}$$

$$4 \cos^2 \frac{2\pi}{3} + 2 \cos \frac{2\pi}{3} - 1 = 0$$

$$\cos \frac{2\pi}{3} = \frac{-2 \pm \sqrt{2^2 - 4 \cdot 4 \cdot -1}}{8}$$

$$= \frac{-2 \pm \sqrt{20}}{8}$$

$$= \frac{-1 \pm \sqrt{5}}{4}$$

$$\text{but } \cos \frac{2\pi}{3} > 0$$

$$\therefore \cos \frac{2\pi}{3} = \frac{-1 + \sqrt{5}}{4}$$

$$d) \text{ let roots be } \alpha, \beta, \alpha\beta$$

$$\text{sum of roots let deriv } 2\alpha + 2\beta = -a \quad (1)$$

$$2\alpha + \beta = 3\alpha\beta + \alpha^2 + \beta^2 = b \quad (2)$$

$$3\alpha + \beta = \alpha\beta(\alpha + \beta) = -c \quad (3)$$

$$\text{from } (1) \text{ and } (2)$$

$$\alpha\beta(-\frac{a}{2}) = -c$$

$$\alpha\beta = \frac{2c}{a}$$

$$\text{from } (3) \quad 3\alpha\beta + \alpha^2 + \beta^2 = b$$

$$3\alpha\beta + (\alpha + \beta)^2 - 2\alpha\beta = b$$

$$\alpha\beta + (\alpha + \beta)^2 = b$$

$$\therefore \frac{2c}{a} + \left(-\frac{a}{2}\right)^2 = b$$

$$\frac{2c}{a} + \frac{a^2}{4} = b$$

$$8c + a^3 = 4ab$$

$$a^3 - 4ab + 8c = 0$$